

# Bill of materials

| ITEM | QTY | REFDES                | VALUE                | MANUFACTURER              | PART NUMBER        | DESCRIPTION                                                                       | CATALOGUE | ORDER CODE            | PACKAGE       |
|------|-----|-----------------------|----------------------|---------------------------|--------------------|-----------------------------------------------------------------------------------|-----------|-----------------------|---------------|
| 1    | 2   | C1,C2                 | 15uF X 25V           | Nichicon                  | UMF1E150MDD        | AluminiumElectrolyEcCapacitors-Radial<br>Leaded Al LyEc Cap 105C Radial FC Series | Mouser    | 647-UMF1E150MDD       | CAP1          |
| 2    | 1   | C3                    | 220uFX4V             | Panasonic                 | ECE-A0GKS221B      | AluminiumElectrolyEcCapacitors-Radial<br>Leaded 4VDC 220uF SU SU BI-POL           | Mouser    | 667-ECE-A0GKS221B     | ECAPR_D62XL25 |
| 3    | 1   | C4                    | 120pF                | MurataElectronics         | RCE5C1H121J0A2H03B | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RCE5C1H121J0A2H3B  | CAP2          |
| 4    | 1   | C5                    | 220pF                | MurataElectronics         | RCER71H221K0A2H03B | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RCER71H221K0A2H03B | CAP2          |
| 5    | 2   | C6,C7                 | 1000pF               | MurataElectronics         | RCE5C1H102J0A2H3B  | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RCE5C1H102J0A2H3B  | CAP2          |
| 6    | 1   | C8                    | 18pF                 | MurataElectronics         | RCE5C1H180J0DBH3A  | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RCE5C1H180J0DBH3A  | CAP2          |
| 7    | 1   | C9                    | 0.1uF                | MurataElectronics         | RHEL81H104K0A2H3B  | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RHEL81H104K0A2H3B  | CAP2          |
| 8    | 1   | C10                   | 0.047uF              | MurataElectronics         | RCER71H473K0A2H3B  | MuIeLayerCeramicCapacitorsMLCC-<br>Leaded 50volts +/-5%                           | Mouser    | 81-RCER71H473K0A2H3B  | CAP2          |
| 9    | 2   | D1,D2                 | 1N4004               | DiodesIncorporated        | 1N4004-T           | RecEfiersVr/400Vlo/1AT/R                                                          | Mouser    | 621-1N4004            | 1N4004        |
| 10   | 2   | D3,D4                 | 1N4148               | ONSemiconductor/Fairchild | 1N4148             | Diodes - General Purpose, Power, Switching<br>100V Io/200mA                       | Mouser    | 512-1N4148            | 1N4148        |
| 11   | 2   | D5,D6                 | 1N5242B              | ONSemiconductor/Fairchild | 1N5242B            | ZenerDiodes12V0.5WZener                                                           | Mouser    | 512-1N5242B           | 1N5242        |
| 12   | 7   | J1,J2,J4,J6,J7,J8,J10 | MKDS1/3-3.81 P<br>HT | PhoenixContact            | 1985836            | FixedTerminalBlocksMKDS1/3-3.81HT                                                 | Mouser    | 651-1985836           | PHOENIX3_381  |
| 13   | 4   | J3,J5,J9,J11          | MKDS1/2-3.81 P<br>HT | PhoenixContact            | 1985823            | FixedTerminalBlocksMKDS1/2-3.81HT                                                 | Mouser    | 651-1985823           | PHOENIX2_381  |
| 14   | 1   | R1                    | 2M2                  | Xicon                     | 271-2.2M-RC        | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-2.2M-RC           | RES1          |
| 15   | 1   | R2                    | 200R                 | Xicon                     | 271-200-RC         | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-200-RC            | RES1          |
| 16   | 1   | R3                    | 5K1                  | Xicon                     | 271-5.1K-RC        | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-5.1K-RC           | RES1          |
| 17   | 1   | R4                    | 10K                  | Xicon                     | 271-10K-RC         | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-10K-RC            | RES1          |
| 18   | 1   | R5                    | 220                  | Xicon                     | 271-220-RC         | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-220-RC            | RES1          |
| 19   | 1   | R6                    | 20K                  | Xicon                     | 271-20K-RC         | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-20K-RC            | RES1          |
| 20   | 2   | R7,R9                 | 3K3                  | Xicon                     | 271-3.3K-RC        | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-3.3K-RC           | RES1          |
| 21   | 1   | R8                    | 1M                   | Xicon                     | 271-1.0M-RC        | MetalFilmResistorsThroughHole1%<br>50PPM                                          | Mouser    | 271-1.0M-RC           | RES1          |
| 22   | 1   | T1                    | EA2622               | CAPIInc. CAPIInc.         | EA2622             | AudioInputTransformer                                                             | CAPI      | EA2622                | EA2622        |
| 23   | 1   | T2                    | EA2503               | AutomatedProcessesINC.    | EA2503             | AudioInputTransformer                                                             | CAPI      | EA2503                | EA2503        |
| 24   | 1   | U1                    | 2520                 |                           | 2520               | AudiooperaEonalamplifier                                                          | MOUSER    | 2520                  | DOA2520       |